

E-Commerce D2C Sales OS

PRODUCT 01 - System Architecture & Technical Build Documentation

1. Executive Overview & System Architecture

The **E-Commerce D2C Sales OS** is a production-grade multi-tenant operating system designed for direct-to-consumer enterprises. It couples an automated fraud/RTO risk engine, multilingual WhatsApp recovery workflows (English, Marathi, Hindi), real-time Tally ERP inventory synchronization, Razorpay signature verification, and Shiprocket automated fulfillment.

Tier	Stack / Tools	Functionality & Production Responsibilities
Frontend	Next.js 15, Tailwind CSS, Lucide Icons, Axios	Multi-tenant subdomain routing, operational dashboards, order and catalog tables
Backend	Node.js, Express, Mongoose, Multer, ExcelJS	REST API microservices, multi-image upload processing, Year 2026 Audit Excel exports
Security	JWT, Bcrypt.js, HMAC-SHA256	Stateless session authentication, password hashing, Razorpay payment verification
Automation	Node-cron, Meta WhatsApp Cloud API, Fast2SMS	Cron schedule (0 8,15,20 * * *) executing [0, 1, 3] days recovery matrix
Integrations	Tally Connector, Shiprocket Logistics, Razorpay	Real-time ERP stock updates, automated AWB generation, digital payment reconciliation

2. Mongoose Schemas & Data Model Specifications

Model	Attributes & Field Types	Constraints & Relationships
Company	<code>_id</code> , <code>name</code> , <code>subdomain</code> , <code>ownerEmail</code> , <code>passwordHash</code> , <code>createdAt</code>	<code>subdomain</code> unique lowercase key; isolates all tenant data.
Product	<code>_id</code> , <code>companyId</code> , <code>title</code> , <code>sku</code> , <code>price</code> , <code>gstPercent</code> , <code>stock</code> , <code>images</code> , <code>tallyItemId</code> , <code>status</code>	<code>sku</code> unique key; <code>companyId</code> ref Company; <code>images</code> up to 5 URLs.
Order	<code>_id</code> , <code>companyId</code> , <code>customerName</code> , <code>phone</code> , <code>items</code> : [{ <code>productId</code> , <code>qty</code> , <code>price</code> }], <code>total</code> , <code>gstTotal</code> , <code>paymentMode</code> , <code>paymentStatus</code> , <code>rtoScore</code> , <code>shiprocketAwb</code> , <code>status</code> , <code>createdAt</code>	<code>paymentMode</code> : COD/UPI/Razorpay; <code>status</code> : valid/packed/shipped/delivered/rto; <code>rtoScore</code> : 0-100.
AbandonedCart	<code>companyId</code> , <code>phone</code> , <code>productId</code> , <code>cartValue</code> , <code>reminderDays</code> , <code>recovered</code> , <code>createdAt</code>	<code>reminderDays</code> : default [0, 1, 3]; tracks recovery status and cron execution.
TallySync	<code>companyId</code> , <code>lastSyncAt</code> , <code>itemsSynced</code> , <code>status</code>	Stores ERP synchronization timestamps and updated stock records.

3. Backend REST API Endpoints Specification

Method & Route	Payload / Query Params	Logic & Execution Result
POST /api/auth/register	name, subdomain, ownerEmail, password	Bcrypt hashes password, creates tenant Company, issues signed JWT token.
POST /api/auth/login	subdomain, email, password	Validates tenant credentials, verifies hash, returns JWT and companyId.
POST /api/products/create	Bearer Auth, FormData (images max 5, title, sku, price, gstPercent, stock)	Multer uploads to /uploads/, stores URLs in DB, computes tax profile.
GET /api/products/list	?companyId=&status;=	Returns tenant catalog sorted by createdAt descending.
POST /api/orders/create	companyId, customerName, phone, items, paymentMode	Computes item subtotal + GST, executes RTO scoring algorithm, creates Order.
GET /api/orders/list	?companyId=&status;=	Returns orders with populated product references (e.g. status=rto).
GET /api/orders/stats	?companyId=	Returns total orders, rtoCount, deliveredCount, packedCount, shippedCount.
POST /api/orders/abandoned/check	companyId, phone, productId, cartValue	Logs pending customer cart with reminder schedule [0, 1, 3] days.
POST /api/orders/shiprocket/ship	orderId	Generates Shiprocket AWB track code, updates order status to 'shipped'.
POST /api/payment/verify	razorpay_order_id, razorpay_payment_id, razorpay_signature, orderId	HMAC SHA256 signature verification; updates order paymentStatus to 'paid'.
POST /api/tally/sync	companyId, tallyData [{sku, stock, price}]	Syncs product inventory stock in bulk, records TallySync log entry.
POST /api/whatsapp/abandoned	abandonedCartId, language (English/Marathi/Hindi)	Dispatches Meta WhatsApp Cloud API message with direct UPI payment link.
GET /api/audit/export	?companyId=&year;=2026	Streams native Excel (.xlsx) file with Title, SKU, Total, GST, Status.

4. Automated Cron Recovery & Multilingual Engine

Cron Schedule: 0 8,15,20 * * * (Runs daily at 08:00 AM, 03:00 PM, and 08:00 PM).
Calculates elapsed days: `diffDays = Math.floor((now - createdAt) / (24 * 60 * 60 * 1000))`.
Triggers WhatsApp + Fast2SMS notifications when `reminderDays.includes(diffDays)`:

- **Day 0:** Immediate high-priority cart recovery message with direct UPI link.
- **Day 1:** 24-hour reminder offering quick checkout before item is released.
- **Day 3:** Final 72-hour scarcity notice before cart cancellation.

Multilingual Messaging (English / Marathi / Hindi Translation Profiles):

Language	Recovery Nudge Format & UPI Deep Link Structure
English	Hello! You left {Item} in your cart. Complete purchase for Rs.{Amount}: upi://pay?pa=merchant@upi&pn;=Store&am;={Amount}&cu;=INR&tn;=CompleteCart
Marathi	Namaskar! Tumchya cart madhil {Item} ajunhi baki ahe. Fakt Rs.{Amount} madhe kharedi purna kara: upi://pay?pa=merchant@upi...
Hindi	Namaste! Aapke cart me {Item} abhi bhi baki hai. Keval Rs.{Amount} me apni kharidari poori karein: upi://pay?pa=merchant@upi...

5. Frontend Next.js Architecture & Views

The frontend is organized under the Next.js App Router with responsive Tailwind CSS components, real-time data fetching through a unified lib/api.js client, and persistent token/tenant caching.

Route / Page	UI Components & Interactive Features
/login	Subdomain, email, and password authentication. Direct link to register tenant, automatic JWT cookie/localStorage sync.
/dashboard	Summary metric cards: Total Products, Total Orders, Abandoned Count, RTO Risk Percentage. Filtered table of high-risk RTO orders with action items.
/products	Product creation form (Title, SKU, Price, GST %, Stock, Multi-file upload) with immediate live gallery and active catalog table.
/orders	Full orders management table (Phone, Total, Payment Mode, Status, RTO Score). 1-click 'Mark Shipped' via Shiprocket and 'Send Invoice WhatsApp' buttons.
/abandoned	Live AbandonedCart monitor table with recovery status pill and immediate 'Send WhatsApp Now' recovery dispatch trigger.
/tally	ERP sync status, live item count sync trigger, and historical Tally audit log viewer.
/settings	Notification toggles (WhatsApp, SMS, UPI AutoPay), language selector (English/Marathi/Hindi), Razorpay/Shiprocket credential inputs, and 'Test Alert' trigger.

6. Run & Done Verification Checklist

Requirement Checklist Item	Implementation Verification Status
✓ Company register/login with subdomain works	Verified via /api/auth/register and /api/auth/login
✓ Product upload with GST calculated works	Verified with Multer multi-file upload and automated 18%/12% tax calc
✓ Product list filtered by companyId works	Verified via /api/products/list?companyId={id}
✓ Dashboard cards correct	Total Products, Total Orders, Abandoned Count, RTO % computed from DB
✓ Order create with RTO score works	Heuristic fraud engine scores COD and customer parameters (0-100)
✓ Abandoned cron sends WhatsApp image + UPI at 0,1,3 days	Configured in cron.js running at 0 8,15,20 * * * with UPI deep links
✓ Tally sync updates stock	POST /api/tally/sync successfully updates catalog stock numbers and logs event
✓ Razorpay verify updates paid	HMAC-SHA256 signature verification updates order status to 'paid'
✓ Excel export downloads	GET /api/audit/export?year=2026 delivers styled .xlsx with Title, SKU, Total, GST, Status
✓ WhatsApp Marathi/Hindi toggle works	Multilingual localized translations generated and delivered dynamically